Boss Challenge — Paper 49 (Class 7)

Acids, Bases & Salts | Reproduction in Plants | Comparing Quantities | Symmetry

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. A substance that tastes sour and turns blue litmus paper red is classed as an:

(A) base
(B) salt
(C) acid
(D) indicator

2. The life process by which living things produce new individuals of their own kind is called:

(A) transpiration
(B) germination
(C) respiration
(D) reproduction

3. A comparison of two quantities of the same kind by division, written with a colon, is called a:

(A) perimeter
(B) ratio
(C) average
(D) percentage

4. A line that divides a figure into two identical halves that fold exactly onto each other is a:

(A) radius
(B) line of symmetry
(C) perimeter
(D) diagonal

5. A substance that feels soapy, tastes bitter and turns red litmus paper blue is a:

(A) salt
(B) neutral substance
(C) acid
(D) base

6. Reproduction in which a new plant arises from a single parent, without flowers, seeds or fusion, is:

(A) fertilisation
(B) sexual reproduction
(C) asexual reproduction
(D) pollination

7. When reduced to lowest terms by dividing out the common factor, the ratio 4 : 6 becomes:

(A) 6 : 4
(B) 2 : 3
(C) 3 : 2
(D) 4 : 6

8. The number of lines of symmetry a square has is:

(A) 8
(B) 4
(C) 1
(D) 2

9. A special substance that tells us whether something is acidic or basic by changing its colour is called an:

(A) indicator
(B) acid
(C) salt
(D) base

10. Growing a new plant from a root, stem or leaf of the parent plant (as in potato or rose) is called:

(A) pollination
(B) vegetative propagation
(C) budding
(D) spore formation

11. The word 'per cent' means 'out of a hundred'. So 7% as a fraction is:

(A) 100/7
(B) 7
(C) 7/100
(D) 7/10

12. Folding a non-square rectangle to make its halves match, you find its lines of symmetry total:

(A) 1
(B) 4
(C) 0
(D) 2

13. Natural litmus, the most common laboratory indicator, is extracted from a plant called:

(A) turmeric
(B) rose
(C) hibiscus
(D) lichen

14. In yeast, a small bulge grows on the parent cell, enlarges and finally separates as a new individual. This is:

(A) budding
(B) spore formation
(C) fertilisation
(D) fragmentation

15. The fraction 1/4 written as a percentage is:

(A) 4%
(B) 40%
(C) 14%
(D) 25%

16. Counting every fold that maps one half onto the other, an equilateral triangle has lines of symmetry totalling:

(A) 1
(B) 3
(C) 0
(D) 2

17. When turmeric paper (a yellow indicator) is touched by a base such as soap solution, it turns:

(A) blue
(B) red
(C) green
(D) colourless

18. Spirogyra, a green water plant, breaks into pieces that each grow into a new plant. This way of reproducing is:

(A) budding
(B) germination
(C) pollination
(D) fragmentation

19. The decimal 0.35 written as a percentage is:

(A) 35%
(B) 3.5%
(C) 350%
(D) 0.35%

20. The number of lines of symmetry a circle has is:

(A) infinitely many
(B) 1
(C) 0
(D) 4

21. The reaction in which an acid and a base cancel each other to give a salt and water is called:

(A) respiration
(B) neutralisation
(C) evaporation
(D) oxidation

22. Ferns and many fungi reproduce by making tiny, light reproductive units that scatter in the air, called:

- **(A)** seeds
- **(B)** spores
- **(C)** gametes
- **(D)** buds

23. Taking twenty per cent of the number one hundred and fifty gives:

- **(A)** 130
- **(B)** 30
- **(C)** 300
- **(D)** 3

24. An isosceles triangle, which has just two equal sides, has this many lines of symmetry:

- **(A)** 2
- **(B)** 0
- **(C)** 1
- **(D)** 3

25. A person with acidity in the stomach is given a mild base to ease the pain. Such a medicine is called an:

- **(A)** an indicator
- **(B)** a salt solution
- **(C)** a vitamin
- **(D)** antacid

26. The male reproductive part of a flower, which produces pollen grains, is the:

- **(A)** stamen
- **(B)** sepal
- **(C)** petal
- **(D)** pistil

27. An article bought for ■400 (cost price) is sold for ■500 (selling price). The profit is:

- **(A)** ■500
- **(B)** ■100
- **(C)** ■25
- **(D)** ■900

28. A scalene triangle, in which all three sides have different lengths, has this many lines of symmetry:

- **(A)** 2
- **(B)** 3
- **(C)** 0
- **(D)** 1

29. An ant's sting injects formic acid into the skin. Rubbing on baking soda (a mild base) helps because it:

- **(A)** adds more acid to the skin
- **(B)** neutralises the acid
- **(C)** cools the skin like ice
- **(D)** washes the dirt away

30. The female reproductive part of a flower, made of stigma, style and ovary, is the:

- **(A)** pistil (carpel)
- **(B)** petal
- **(C)** stamen
- **(D)** filament

31. An item costing ■250 is sold for ■200. This is a:

- **(A)** loss of ■50
- **(B)** profit of ■50
- **(C)** no profit and no loss
- **(D)** loss of ■450

32. The capital letter that has a single VERTICAL line of symmetry is:

- **(A)** F
- **(B)** P
- **(C)** A
- **(D)** R

33. A field whose soil has become too acidic for crops is treated by the farmer with:

- **(A)** plain sand
- **(B)** a base such as lime (quicklime)
- **(C)** table salt
- **(D)** more acid such as vinegar

34. Moving pollen grains from a flower's anther onto a stigma, the step before fertilisation, is named:

- **(A)** pollination
- **(B)** dispersal
- **(C)** germination
- **(D)** fertilisation

35. A shopkeeper buys a toy for ■200 and sells it at a 10% profit. The selling price is:

- **(A)** ■2000
- **(B)** ■180
- **(C)** ■210
- **(D)** ■220

36. The capital letter B has a line of symmetry that is:

- **(A)** vertical
- **(B)** diagonal
- **(C)** horizontal
- **(D)** both vertical and horizontal

37. The sour taste of lemons, oranges and other citrus fruits is due to the presence of:

- **(A)** acetic acid
- **(B)** citric acid
- **(C)** lactic acid
- **(D)** hydrochloric acid

38. When pollen from a flower's anther reaches the stigma of a DIFFERENT flower, the pollination is called:

- **(A)** cross-pollination
- **(B)** fertilisation
- **(C)** germination
- **(D)** self-pollination

39. A sum of ■1000 is kept for 2 years at 5% per year. The simple interest earned is:

- **(A)** ■1100
- **(B)** ■100
- **(C)** ■50
- **(D)** ■200

40. Turning a figure about a fixed centre point so that it looks exactly the same is called:

- **(A)** translation
- **(B)** rotational symmetry
- **(C)** reflection
- **(D)** line symmetry

41. Vinegar, used in pickles and cooking, gets its sharp sour taste from:

- **(A)** acetic acid
- **(B)** formic acid
- **(C)** tartaric acid
- **(D)** citric acid

42. The fusion of the male gamete from a pollen grain with the female gamete (egg) inside the ovule is called:

- **(A)** germination
- **(B)** pollination
- **(C)** budding
- **(D)** fertilisation

43. A water sample is 30% salt by mass. In 500 g of this water, the mass of salt is:

- **(A)** 350 g
- **(B)** 30 g
- **(C)** 150 g
- **(D)** 15000 g

44. The order of rotational symmetry of a square (how many times it matches itself in one full turn) is:

- **(A)** 2
- **(B)** 4
- **(C)** 8
- **(D)** 1

45. Curd and sour milk taste sour because, as they ferment, bacteria produce:

- **(A)** acetic acid
- **(B)** hydrochloric acid
- **(C)** citric acid
- **(D)** lactic acid

46. After fertilisation, the ovule of a flower develops into a seed, while the ovary grows into the:

- **(A)** fruit
- **(B)** flower
- **(C)** root
- **(D)** leaf

47. A shirt marked ■800 is sold at a 25% discount. The discount amount is:

- **(A)** ■200
- **(B)** ■25
- **(C)** ■775
- **(D)** ■600

48. For a shape with rotational symmetry of order n, the smallest angle of rotation that maps it onto itself is:

- **(A)** 180° / n
- **(B)** 360° x n
- **(C)** n / 360°
- **(D)** 360° / n

49. The strong acid that our own stomach makes to help digest food and kill germs is:

- **(A)** hydrochloric acid
- **(B)** acetic acid
- **(C)** citric acid
- **(D)** lactic acid

50. Light seeds with wings or hair, like those of the drumstick or dandelion, are carried away from the parent mainly by:

(A) wind
(B) animals
(C) bursting of the fruit
(D) water

51. The ratio 3 : 5 expressed as a percentage of the first quantity to the total is:

(A) 60%
(B) 3%
(C) 37.5%
(D) 30%

52. An equilateral triangle matches itself on turning every:

(A) 60°
(B) 90°
(C) 120°
(D) 360°

53. Lime water, used to test for carbon dioxide, is a solution of the base:

(A) citric acid
(B) hydrochloric acid
(C) calcium hydroxide
(D) sodium chloride

54. Of 200 seeds sown in a tray, 150 sprout into seedlings. The germination percentage is:

(A) 75%
(B) 50%
(C) 150%
(D) 25%

55. If 5 pens cost ■60, then at the same rate the cost of 8 pens is:

(A) ■75
(B) ■480
(C) ■96
(D) ■68

56. A regular pentagon (5 equal sides) has lines of symmetry numbering:

(A) 5
(B) 10
(C) 2
(D) 1

57. Which of these is a NEUTRAL substance that changes the colour of neither red nor blue litmus?

(A) vinegar
(B) soap solution
(C) common salt solution
(D) lemon juice

58. A flower that has both stamens and a pistil - the male and female parts together - is called a:

(A) female flower
(B) male flower
(C) unisexual flower
(D) bisexual flower

59. In a class of 40 students, 60% are girls. The number of boys in the class is:

(A) 24
(B) 16
(C) 4
(D) 40

60. A flower has 5 identical petals spaced evenly around its centre. Its order of rotational symmetry is:

(A) 5
(B) 2
(C) 10
(D) 1

61. Factory waste water is often acidic. Before it is let into a river, it is first treated with a base to:

(A) make it more acidic
(B) neutralise it and make it safe
(C) turn it into a salt crystal
(D) colour it with an indicator

62. In a garden bed there are 60 marigold plants. If 20% of them die from a disease, the number of plants that die is:

(A) 12
(B) 48
(C) 20
(D) 40

63. The price of a bag rose from ■500 to ■600. The percentage increase in price is:

(A) 100%
(B) 20%
(C) 10%
(D) 16.7%

64. Because a regular polygon's lines of symmetry equal its sides, a regular six-sided hexagon has:

- **(A)** 12
- **(B)** 4
- **(C)** 6
- **(D)** 3

65. Toothpastes are usually mildly basic. They protect teeth by neutralising the acids that:

- **(A)** the toothbrush bristles release
- **(B)** the toothpaste itself contains
- **(C)** flow in from drinking water
- **(D)** bacteria make from leftover food

66. The 'eyes' on a potato are buds that can sprout into new plants. Growing potatoes from these eyes is an example of:

- **(A)** seed germination
- **(B)** spore formation
- **(C)** pollination
- **(D)** vegetative propagation

67. A test is marked out of 80. A student scores 60. As a percentage, the score is:

- **(A)** 80%
- **(B)** 20%
- **(C)** 75%
- **(D)** 60%

68. A leaf is often symmetric along its midrib, so a typical simple leaf has this many lines of symmetry:

- **(A)** infinitely many
- **(B)** 0
- **(C)** 2
- **(D)** 1

69. A 200 g sample of dilute acid solution is 15% acid by mass. The mass of pure acid in it is:

- **(A)** 15 g
- **(B)** 185 g
- **(C)** 30 g
- **(D)** 3000 g

70. A Bryophyllum leaf placed on moist soil sprouts tiny new plants along its notched edges. This shows reproduction by:

- **(A)** pollen grains
- **(B)** spores
- **(C)** leaves (vegetative propagation)
- **(D)** seeds

71. Two quantities are in the ratio 1 : 4 and together they total 250 mL. The smaller quantity is:

(A) 50 mL
(B) 200 mL
(C) 62.5 mL
(D) 125 mL

72. A figure that looks the same after a half turn (180°) but not after a quarter turn has rotational symmetry of order:

(A) 0
(B) 2
(C) 4
(D) 1

73. In a neutralisation, acid and base are mixed in the ratio 2 : 3. To use 40 mL of acid, the base needed is:

(A) 26.7 mL
(B) 120 mL
(C) 60 mL
(D) 40 mL

74. In a field the ratio of male (pollen-bearing) flowers to female flowers on the plants is 3 : 2. If there are 30 male flowers, the female flowers number:

(A) 30
(B) 45
(C) 12
(D) 20

75. A solution is made by mixing acid and water in the ratio 1 : 3. The percentage of acid in the solution is:

(A) 75%
(B) 13%
(C) 25%
(D) 33.3%

76. A parallelogram (that is not a rectangle or rhombus) has lines of symmetry numbering:

(A) 4
(B) 2
(C) 1
(D) 0

77. Of 250 liquid samples tested with litmus, 40% turned blue litmus red. The number of acidic samples is:

(A) 40
(B) 210
(C) 150
(D) 100

78. A coconut can float across the sea and sprout on a far shore. Its seed is mainly dispersed by:

- **(A)** wind
- **(B)** bursting of the fruit
- **(C)** water
- **(D)** animals

79. An item is sold for ■360 at a 20% profit. Its cost price was:

- **(A)** ■340
- **(B)** ■288
- **(C)** ■300
- **(D)** ■432

80. When a shape is flipped across a line to give its mirror image, the result is its:

- **(A)** translation
- **(B)** enlargement
- **(C)** rotation
- **(D)** reflection

81. The salt formed when hydrochloric acid is exactly neutralised by the base sodium hydroxide is:

- **(A)** sodium chloride (common salt)
- **(B)** sodium hydroxide
- **(C)** calcium carbonate
- **(D)** hydrochloric acid

82. A money plant or rose is commonly grown by planting a piece of its stem in soil. This piece is called a:

- **(A)** gamete
- **(B)** seed
- **(C)** spore
- **(D)** cutting

83. A football team played 25 matches and won 60% of them. The number of matches won is:

- **(A)** 15
- **(B)** 6
- **(C)** 10
- **(D)** 60

84. A regular octagon (8 equal sides) has rotational symmetry of order:

- **(A)** 4
- **(B)** 8
- **(C)** 16
- **(D)** 2

85. China rose indicator turns dark pink (magenta) in acids and green in bases. Dipped in soap solution, it turns:

- **(A)** colourless
- **(B)** green
- **(C)** blue
- **(D)** dark pink (magenta)

86. Out of 400 seeds tested, 70% germinate. The number of seeds that did NOT germinate is:

- **(A)** 70
- **(B)** 30
- **(C)** 280
- **(D)** 120

87. Out of every 100 g of a snack, 12 g is sugar. In a 250 g pack, the mass of sugar is:

- **(A)** 48 g
- **(B)** 12 g
- **(C)** 30 g
- **(D)** 3 g

88. The English capital letter that has BOTH a vertical and a horizontal line of symmetry is:

- **(A)** P
- **(B)** B
- **(C)** H
- **(D)** A

89. A bottle reads '25% acid by mass'. To get 50 g of pure acid, the total mass of solution you must take is:

- **(A)** 1250 g
- **(B)** 75 g
- **(C)** 12.5 g
- **(D)** 200 g

90. Why does a flowering plant produce far more pollen grains and seeds than the number of new plants that finally grow?

- **(A)** many are lost, so extra numbers improve the chance some survive
- **(B)** more seeds always means heavier fruit only
- **(C)** plants cannot control how many they make
- **(D)** the extra pollen and seeds are simply waste with no purpose

91. The ratio of the length to the breadth of a field is 5 : 2. If the length is 35 m, the breadth is:

- **(A)** 70 m
- **(B)** 17.5 m
- **(C)** 87.5 m
- **(D)** 14 m

92. Every regular polygon has its number of lines of symmetry exactly equal to its number of:

- **(A)** diagonals
- **(B)** right angles
- **(C)** sides
- **(D)** vertices it shares with a circle

93. Blue litmus is dipped into four liquids. It stays blue in all but one, where it turns red. That liquid is:

- **(A)** the most dilute one
- **(B)** the only acidic one
- **(C)** the only basic one
- **(D)** the only neutral one

94. Burr seeds with tiny hooks that cling to an animal's fur and are carried far away are dispersed by:

- **(A)** animals
- **(B)** wind
- **(C)** water
- **(D)** bursting of the fruit

95. A number is increased to 130% of itself. If the original number was 40, the new number is:

- **(A)** 130
- **(B)** 52
- **(C)** 70
- **(D)** 13

96. A windmill design with 4 identical blades evenly placed looks the same on turning by 90° but has no line of symmetry. It has:

- **(A)** no symmetry at all
- **(B)** line symmetry only
- **(C)** both line and rotational symmetry
- **(D)** rotational symmetry only

97. Two acidic solutions are compared. Solution X is 10% acid by mass and Solution Y is 25% acid by mass. The stronger acid solution is:

- **(A)** Solution Y
- **(B)** both are equal
- **(C)** neither can be compared
- **(D)** Solution X

98. The sticky tip of the pistil, on which pollen grains land during pollination, is called the:

- **(A)** filament
- **(B)** anther
- **(C)** stigma
- **(D)** ovary

99. A garden has roses and lilies in the ratio 7 : 3. The percentage of the flowers that are lilies is:

(A) 70%

(B) 43%

(C) 3%

(D) 30%

100. A rhombus (a slanted shape with all four sides equal) has lines of symmetry numbering:

(A) 2

(B) 0

(C) 4

(D) 1